

2025 Atlantic Sea Scallop (*Placopecten magellanicus*) Research Track Working Paper: Thermal Habitat

An exploration of thermal habitat changes and risk in the Mid-Atlantic Bight using GLORYS, 1993-2023

Joseph Caracappa, Deborah Hart, Jui-Han Chang, Kimberly Hyde

Abstract

Surveys indicate a gradual truncation of the southern and inshore range of Atlantic sea scallop in the Mid-Atlantic Bight (MAB) and increasing adult mortality in these areas. To help quantify retrospective changes in thermal habitat with respect to sea scallop physiology, an examination of GLORYS12V1 Global Ocean Physics Reanalysis was used to develop indicators of thermal habitat and heat stress events in the MAB. These analyses showed a decrease in baseline thermal habitat from 1993 to 2023 with a shift in suitable habitat offshore. The spatial extent and duration of heat stress events also increased over this time, particularly in September and October. There was no significant trend in the intensity of heat stress events. Heat stress events were detected in proximity to the 2022 mortality event in Elephant Trunk, but other heat stress events were observed that did not result in mortality. In order to build more robust indicators of mortality events, further analysis will focus on assessing additive stress of food availability and other environmental variables as well as a focus on the timing of heat stress events.

1. Introduction

Truncations in the southern and inshore ranges of the sea scallop (*Placopecten magellanicus*) in the Mid-Atlantic Bight (MAB) have been observed since the early 2000s (Figure 1.1). Typically, recruitment is still observed in these areas, but survival past age two is low. One example of this occurred between 2022 and 2023 in the Delmarva and Elephant Trunk areas off of Maryland and Delaware (Figure 1.2). Above average recruitment was observed in both of these areas on the 2022 surveys. This cohort was also observed during the 2022 Northeast Fisheries Science Center (NEFSC) fall trawl survey, which surveyed this area in September 2022. However, few of these scallops remained when the area was surveyed again the next spring by the 2023 dredge and Habcam surveys. The most likely explanation for the mortality event is that it was due to high temperatures; maximum temperatures in these areas occur typically in October when stratification starts to break down.

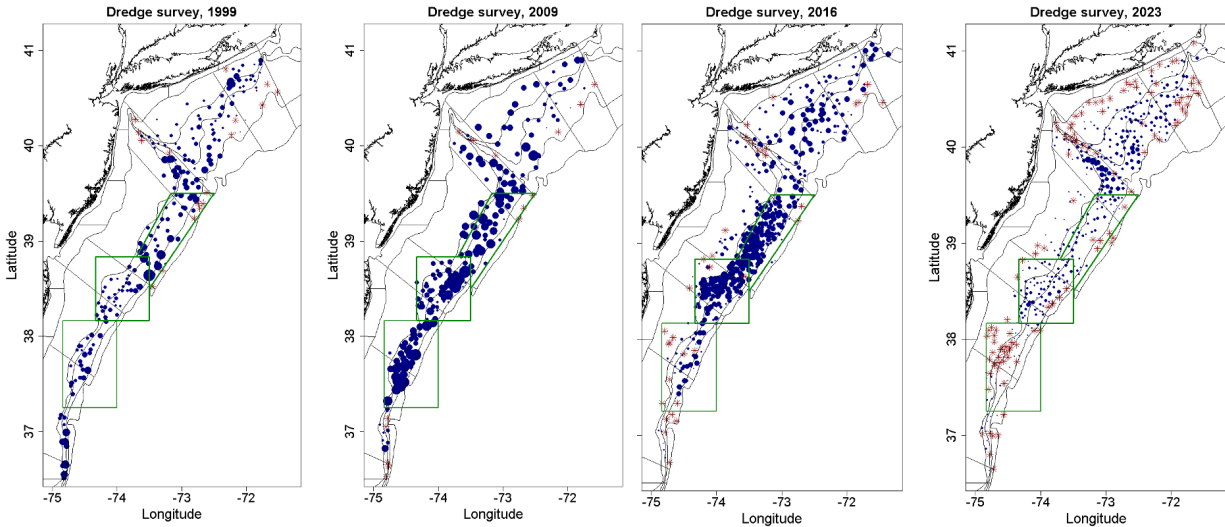


Figure 1.1: Mid-Atlantic dredge surveys from 1999, 2009, 2016, and 2023. Numbers caught per tow are indicated by the size of the blue dots; red stars are locations where no scallops were caught. These figures document a clear decline in scallop abundance along the southern and inshore portions of the Mid-Atlantic.

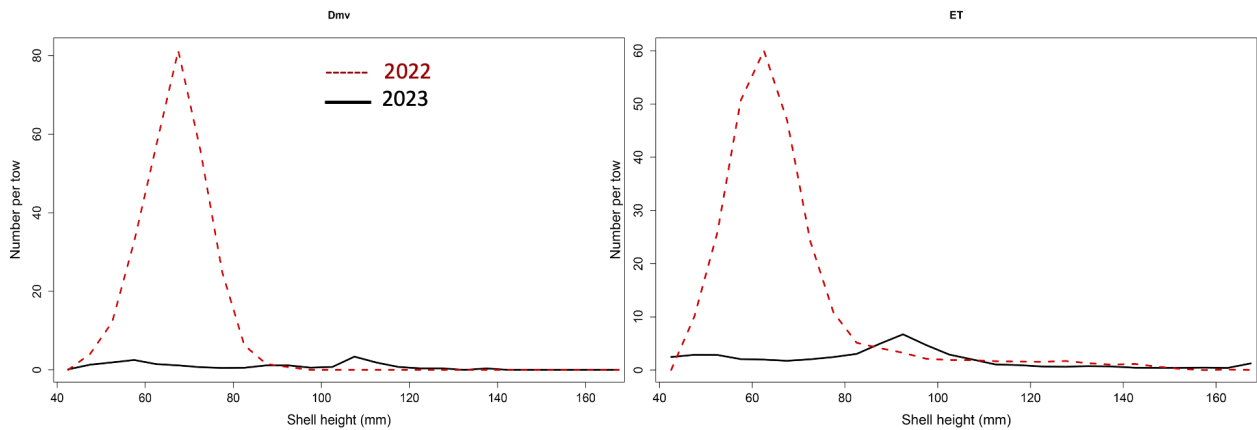


Figure 1.2: Mean stratified number of scallops by shell height in Delmarva (left) and Elephant Trunk (right) in 2022-2023 from the scallop dredge survey. A substantial two year old year class was observed in both areas during the 2022 survey (dashed red line) at around 60 mm. Very few of these were observed during the 2023 survey (black line).

The objective of this working paper is to develop thermal habitat indices for Atlantic sea scallops to better quantify heat stress and heat-related mortality events. The general approach was to assess bottom temperature patterns in the MAB using GLORYS12V1 (CMEMS, 2018) reanalysis with respect to experimental thermal tolerance limits. Then additional analyses were conducted to quantify the frequency, intensity, and extent of the scallop-specific heat stress events.

In the Mid-Atlantic, *Placopecten magellanicus* is mostly found in a narrow depth range between 35 and 80 m. The offshore extent is mainly limited by the spat predator *Astropecten americanus* (Hart 2006, Shank et al. 2012), while thermal stress is likely determining its southern and inshore extent in the Mid-Atlantic, which constitutes the southern end of the sea scallop's range. Limited experimental studies on *P. magellanicus* thermal tolerance indicates that temperatures exceeding 18-19°C result in physiological stress (Frenette, 2004; Guderley et al., 2009) and temperatures above 23°C are immediately lethal (Frenette & Parsons, 2000). Food supply (mainly phytoplankton) likely interacts with temperature; higher phytoplankton levels improve the scallops' ability to tolerate high temperatures (Zang et al., 2023). Dissolved oxygen may be another interacting factor.

In situ bottom temperature is sparsely available over the MAB shelf. These sources include temperature sensors on some survey gear, thermal profiles at fixed stations seasonally from NEFSC's Ecological Monitoring (EcoMon) surveys, and temperature sensors on lobster traps (eMOLT). Even together, these data products do not produce the complete spatial or temporal coverage in the MAB needed to evaluate thermal stress in sea scallops within different management regions. In contrast, GLORYS is a global reanalysis oceanographic model product that provides bottom temperature (amongst other oceanographic metrics) at a 1/12° spatial and daily temporal resolution. As a reanalysis product, GLORYS' oceanographic processes are nudged towards observations (including those from NEFSC's surveys and EcoMon, satellite data, and the World Ocean Database) using a reduced-order Kalman filter. Though not as reliable as *in situ* observations, GLORYS provides bottom temperature at the spatial and temporal resolution needed to investigate thermal habitat for sea scallops while still being grounded in observations.

2. Methods

2.1. GLORYS data access and processing

Daily bottom temperature data were obtained from GLORYS using Copernicus Marine Services API toolbox. The Scallop Area Management Simulation (SAMS) areas were used as the spatial extent for these analyses, which extend from 35°N to 41.5°N where GLORYS bathymetry was 15 to 80m. All GLORYS grid cells that entirely or partially overlap with the areas modeled by the Scallop Area Management Simulator forecasting model (referred hereafter as SAMS areas) were extracted for further analysis. This resulted in a spatial extent from GLORYS of 4.3×10^4 km² compared to the SAMS areas of 3.6×10^4 . The increased extent of the GLORYS data is due to the area of 1/12° grid cells overlapping the SAMS areas.

2.2. Baseline thermal habitat estimates

Based on a literature review and unpublished experiments on sea scallop thermal tolerance, a critical thermal threshold of 18°C was chosen as the point at which heat-stress conditions begin. First, a monthly mean and maximum bottom temperature was calculated for each GLORYS grid

cell from 1993 to 2023. Then, the v (baseline thermal habitat, BTH) was defined by the cells where the maximum monthly bottom temperature never exceeds 18°C. This BTH represents areas within the SAMS areas where sea scallops are unlikely to experience thermal stress. Additionally, an unsuitable thermal habitat (UTH) area was defined as areas where the mean monthly bottom temperature exceeded 18°C. These UTH areas describe regions within SAMS areas where sea scallops would typically experience thermal stress. Core thermal habitats and thermally uninhabitable areas were also calculated across all years 1993 to 2023, as long-term regional characteristics.

2.3. Characteristics of heat stress events

Heat stress events (HSE) were broadly defined for a location if bottom temperatures exceeded 18°C any GLORYS grid cell. However, three main factors were considered in characterizing HSE: spatial extent, duration, and intensity. Measures of spatial extent intend to inform where such events are occurring as well how widespread they are over a given area. However, given that sea scallops are not uniformly distributed within the SAMS areas, spatial extent indices need to be informed by population distribution information to identify potential impacts to scallops. Duration refers to how long heat stress conditions persist at a given location. Short exposures are potentially less impactful than longer ones at a given temperature. Lastly, intensity is a measure of the severity of the event, where even for brief events, sufficiently high temperatures can result in mortality. All three factors are needed to assess the impact of a HSE on sea scallops. Long-term exposure at mildly stressful temperatures could have similar effects to short but intense events. Additionally, even prolonged high intensity events may not have an impact on the fishery if the sea scallop density at that location is low.

The spatial extent of HSEs was defined as the proportion of the total SAMS area that exceeded 18°C. This was performed both on the daily GLORYS bottom temperature as well as the monthly maximum temperature for each grid cell. The duration of heat stress event (HSD) was defined as the number of days in a year where a GLORYS cell was greater than 18°C within the fall (September through November). Scallops spawn in the fall within the MAB, which is also when bottom temperatures are at their peak, making them increasingly vulnerable to HSEs. Lastly, heat stress intensity (HSI) was defined as the degree days for days over 18°C, where an intensity of 0 indicates no days exceeding 18°C. HSI was calculated for each GLORYS grid cell using daily data, and was also summed across the entire SAMS area to get a regional HSI.

3. Results

3.1. Thermal habitat characterization

Mean conditions from the entire GLORYS hindcast (1993 to 2023) show that maximum monthly temperature (BTH) was suitable throughout most of the SAMS area throughout each year. The entire SAMS area was considered part of sea scallops BTH in January through June (Figure 1.3A and 1.3B). In July and August, a small area off of Long Island is excluded which expands into the New York Bight and the southern MAB in September and October (Figure 1.3C and

1.3D) . In November small areas off of Virginia are excluded from BTH, with all areas being suitable in December (Figure 1D).

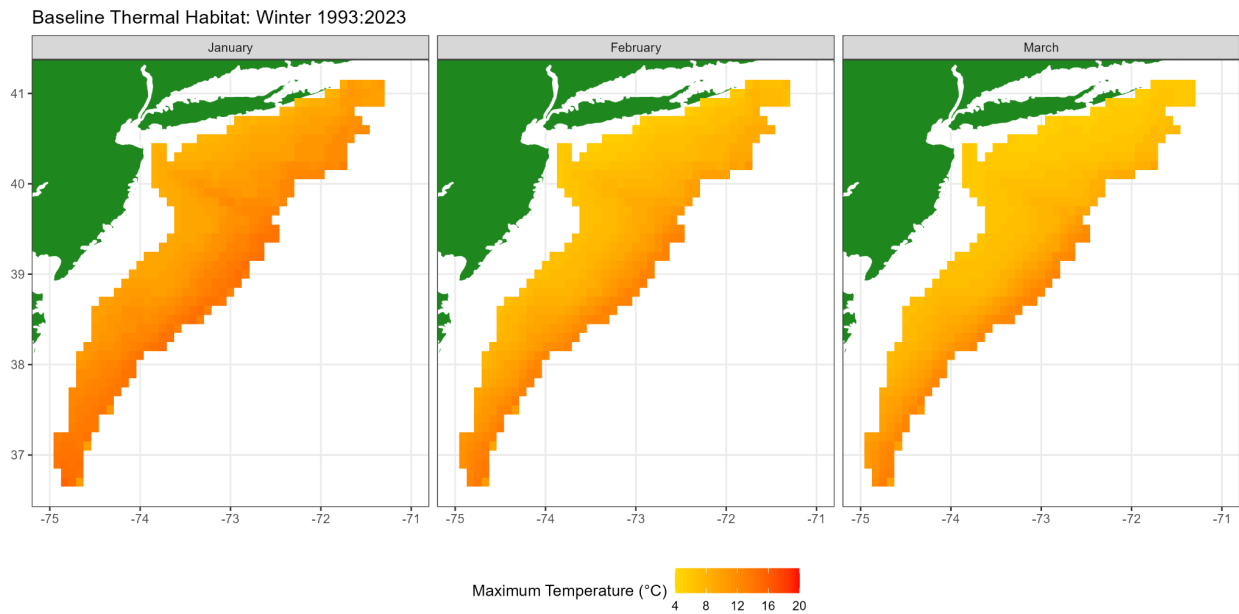


Figure 1.3A: Baseline thermal habitat based on winter (January - March) mean monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas not within baseline thermal habitat.

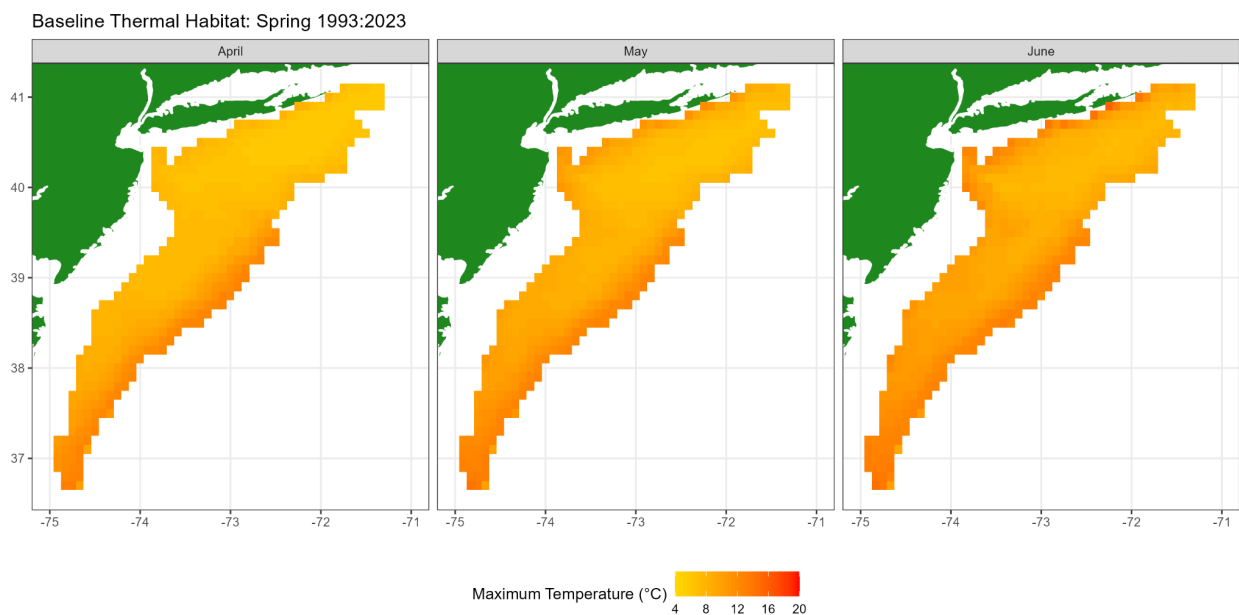


Figure 1.3B: Baseline thermal habitat based on spring (April - June) mean monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas not within

baseline thermal habitat.

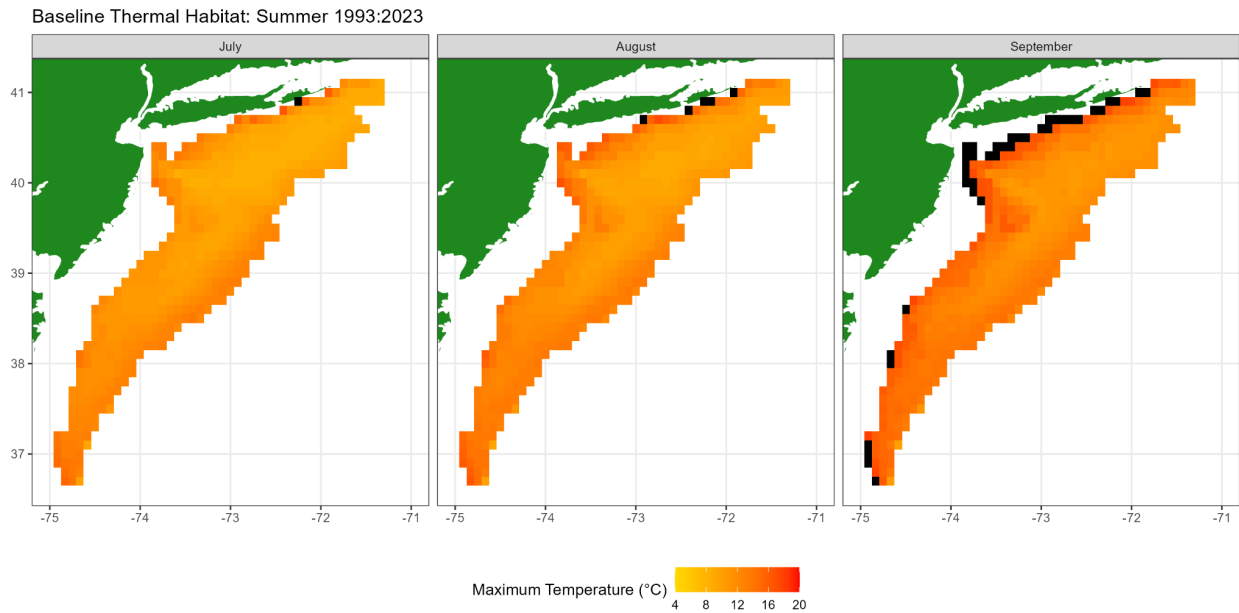


Figure 1.3C: Baseline thermal habitat based on summer (July - September) mean monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas not within baseline thermal habitat.

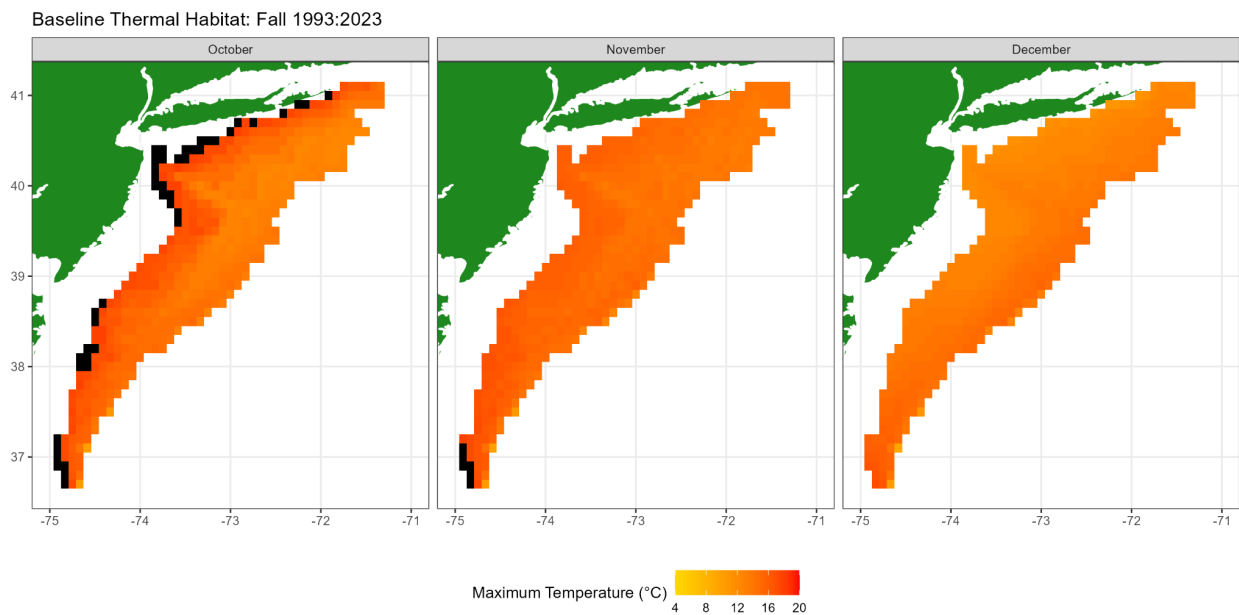


Figure 1.3D: Baseline thermal habitat based on fall (October - December) mean monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas not within baseline thermal habitat.

To describe how BTH has changed over time, decadal monthly means were calculated using the same metrics as above. Nearly the entire SAMS area was considered BTH from 1993 to 2002 (Figure 1.4A) with small exceptions off Long Island in February and Long Island and the New York Bight in September. This pattern was identical from 2003 to 2012 (Figure 1.4B). Most of the expansion of areas outside BTH occurs in the most recent decade (2013 to 2022; Figure 1.4C). In the most recent decade (2013 to 2022), most of the inshore SAMS area is UTH in September and October (Figure 1.4C).

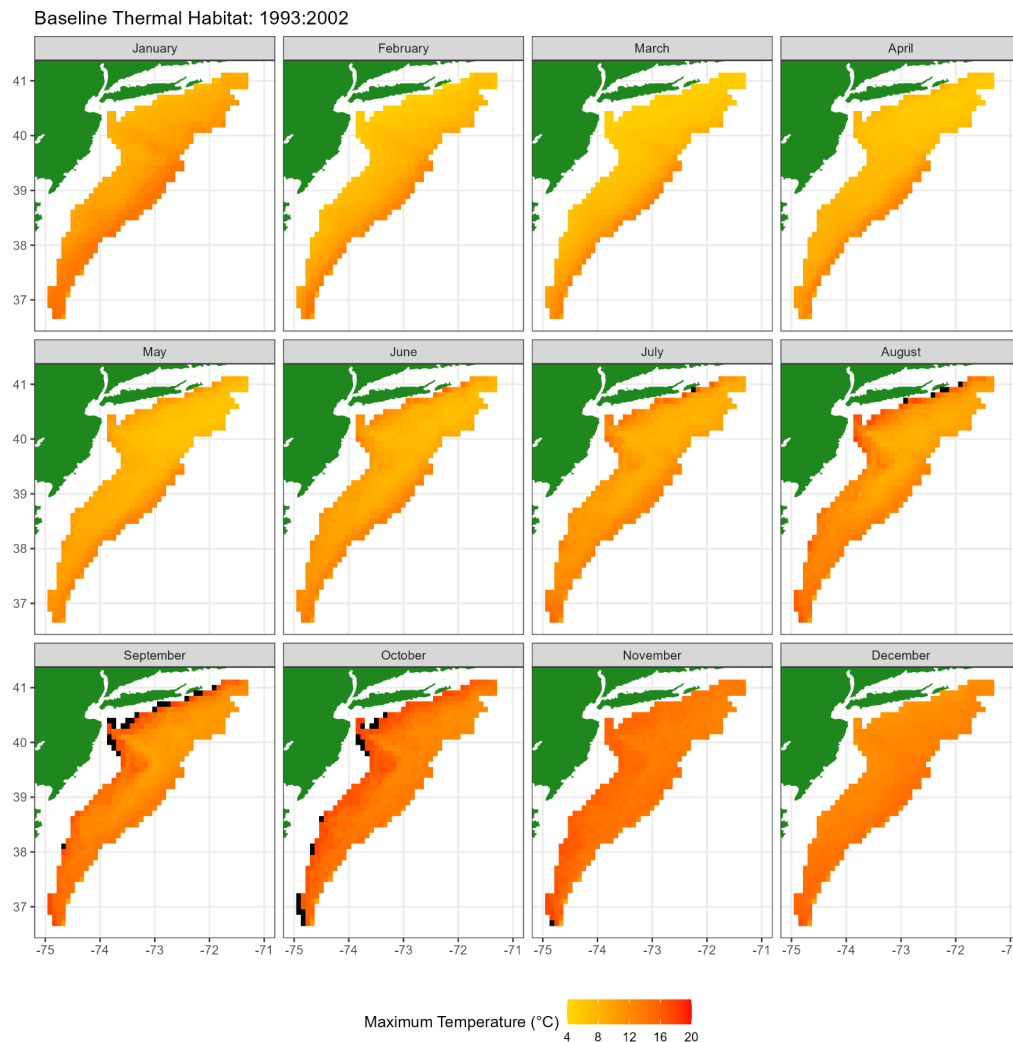


Figure 1.4A: Decadal baseline thermal habitat based mean monthly bottom temperature from 1993 to 2002 using GLORYS reanalysis. Black cells indicate areas not within baseline thermal habitat.

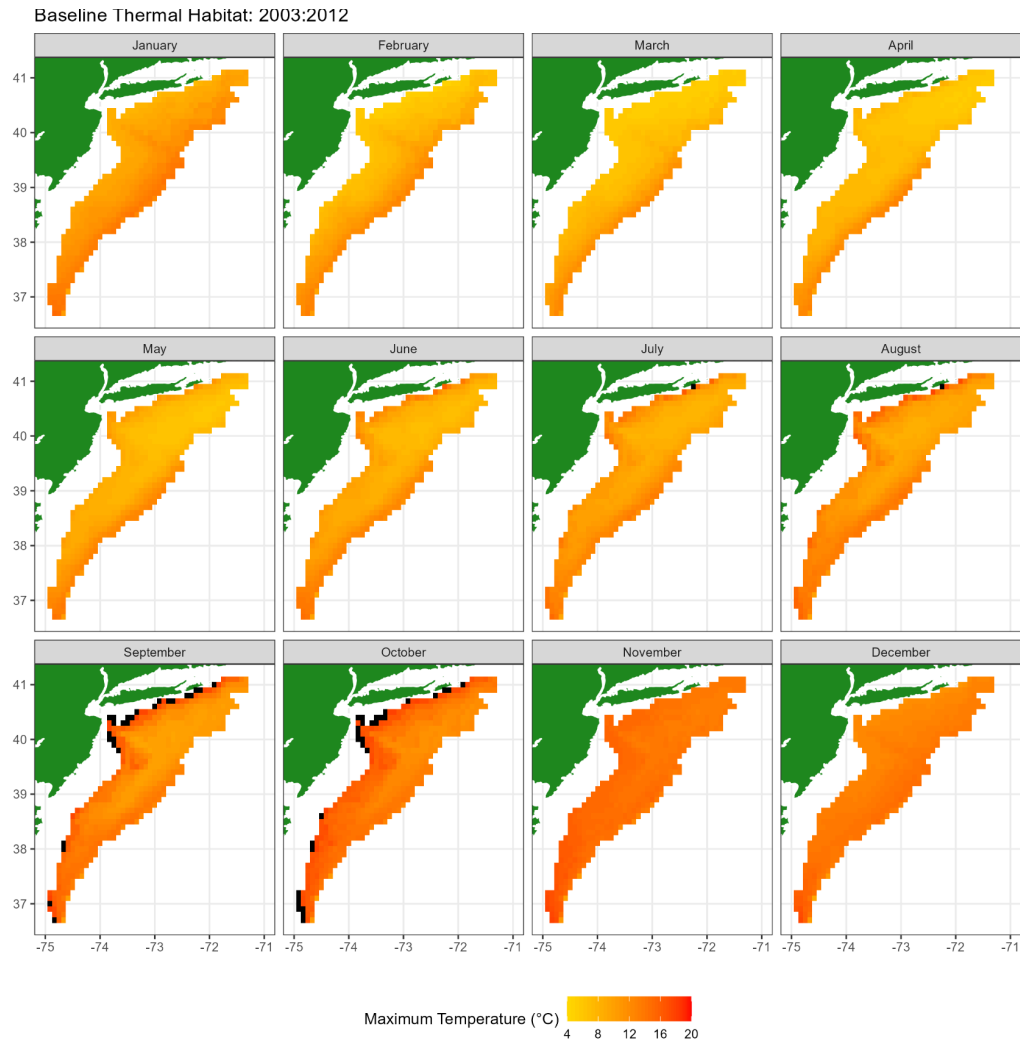


Figure 1.4B: Decadal baseline thermal habitat based mean monthly bottom temperature from 2003 to 2012 using GLORYS reanalysis. Black cells indicate areas not within baseline thermal habitat.

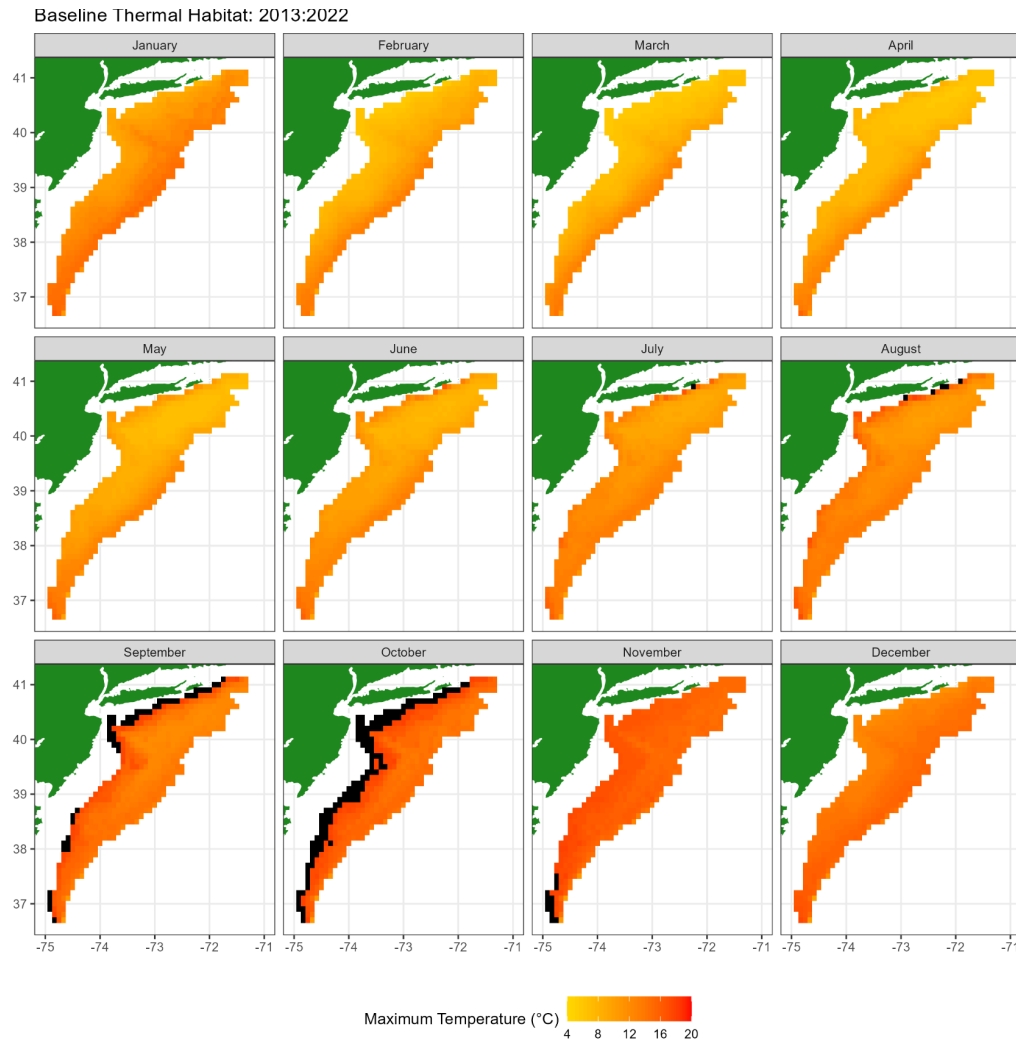


Figure 1.4C: Decadal baseline thermal habitat based mean monthly bottom temperature from 2013 to 2022 using GLORYS reanalysis. Black cells indicate areas not within baseline thermal habitat.

These long-term spatial and seasonal patterns for BTH are similar for UTH but with an increased spatial extent. No areas are considered UTH from January through August (Figures 1.5A, 1.5B, and 1.5C). In September, small areas of UTH exist off of Long Island and New Jersey (Figure 1.5C). From October through December, no parts of the region are UTH (Figure 1.5D).

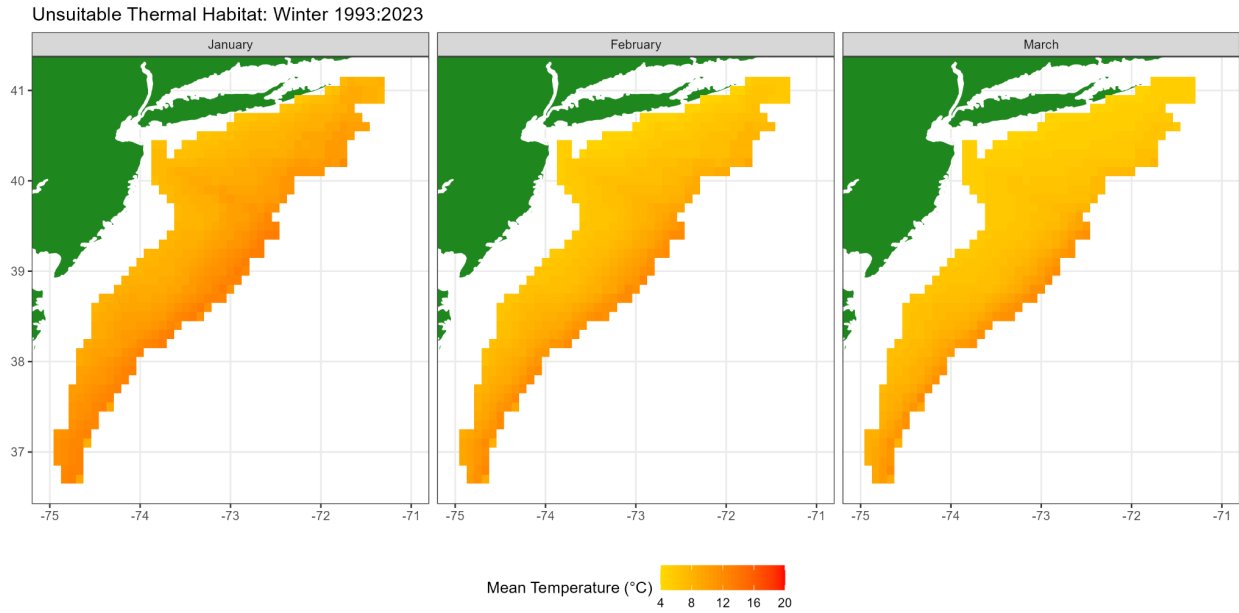


Figure 1.5A: Unsuitable thermal habitat based on winter (January - March) maximum monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas with unsuitable thermal habitat.

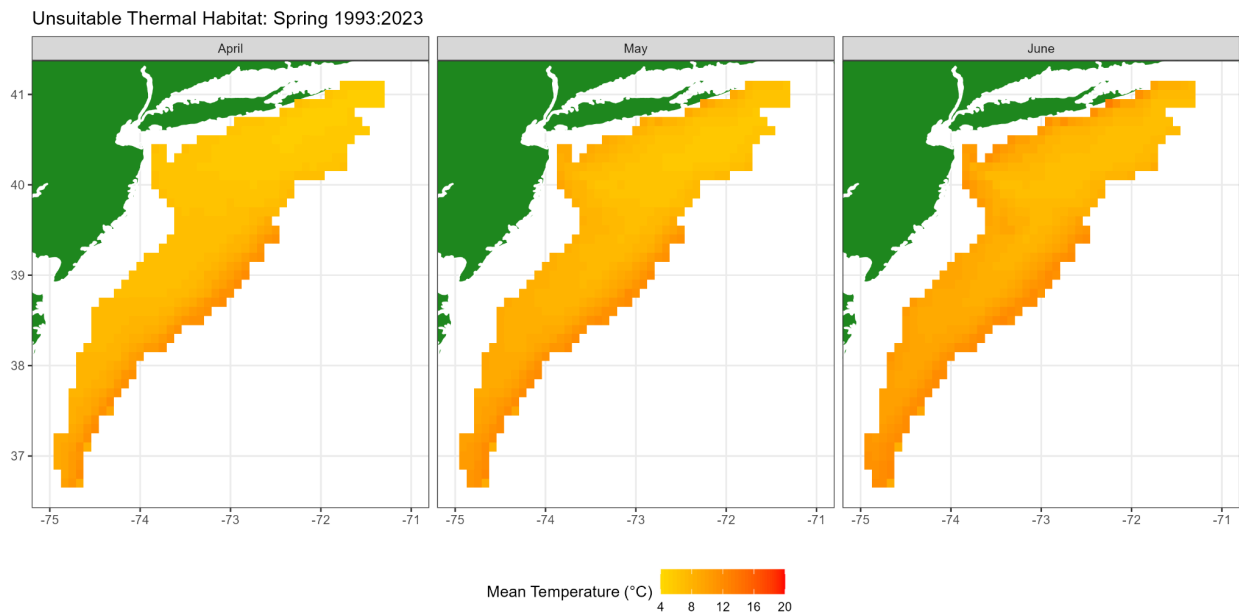


Figure 1.5B: Unsuitable thermal habitat based on spring (April - June) maximum monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas with unsuitable thermal habitat.

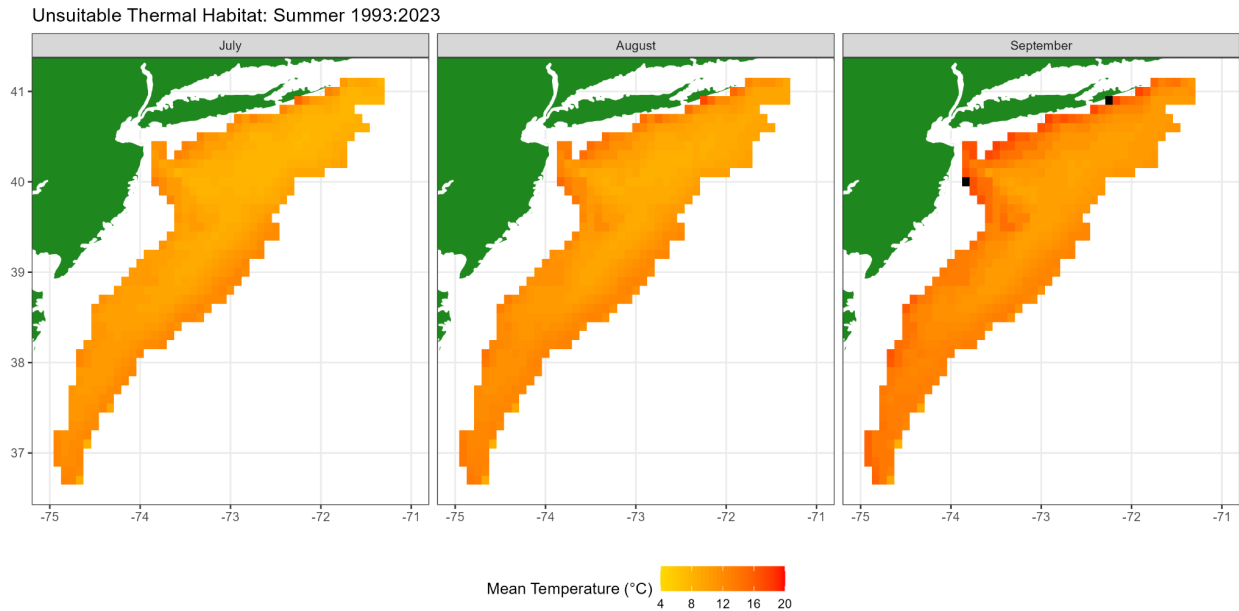


Figure 1.5C: Unsuitable thermal habitat based on summer (July - September) maximum monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas with unsuitable thermal habitat.

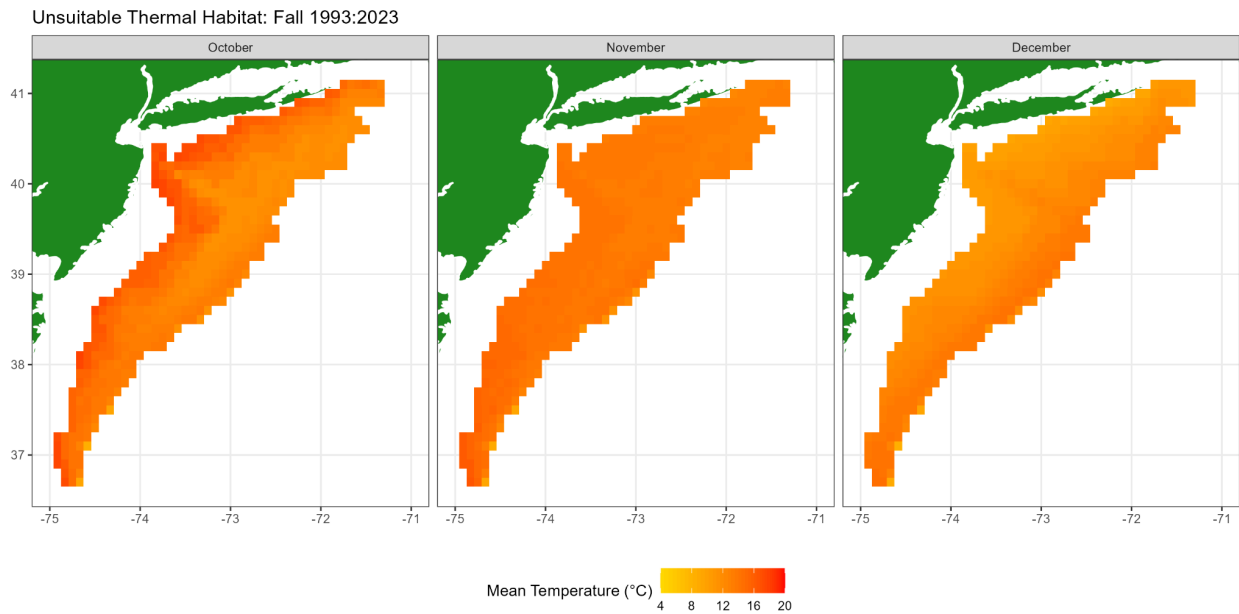


Figure 1.5D: Unsuitable thermal habitat based on fall (October - December) maximum monthly bottom temperature from 1993 to 2023 using GLORYS reanalysis. Black cells indicate areas with unsuitable thermal habitat.

As with BTH, an analysis of decadal UTH shows similar patterns where an increase in UTH is present along the inshore SAMS areas in recent years. From 1993 to 2002, relatively small regions of UTH existed off Long Island and the New York Bight in February and September

(Figure 1.6A). This pattern persists from 2003 to 2012 (Figure 1.6B). In the most recent decade (2013 to 2022), larger areas off of Long Island, the New York Bight, and the southern MAB are UTH in September and October (Figure 1.6C).

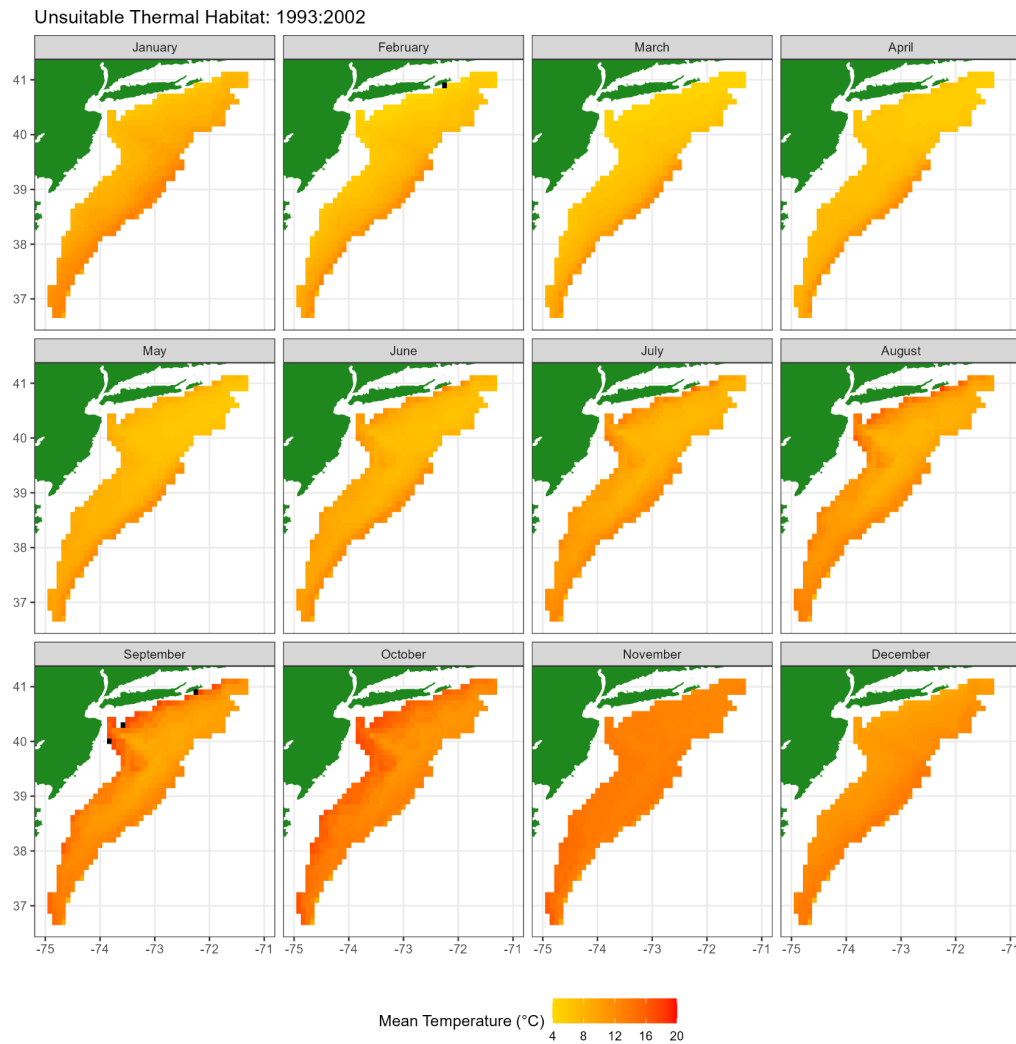


Figure 1.6A: Decadal unsuitable thermal habitat based on maximum monthly bottom temperature from 1993 to 2002 using GLORYS reanalysis. Black cells indicate areas with unsuitable thermal habitat.

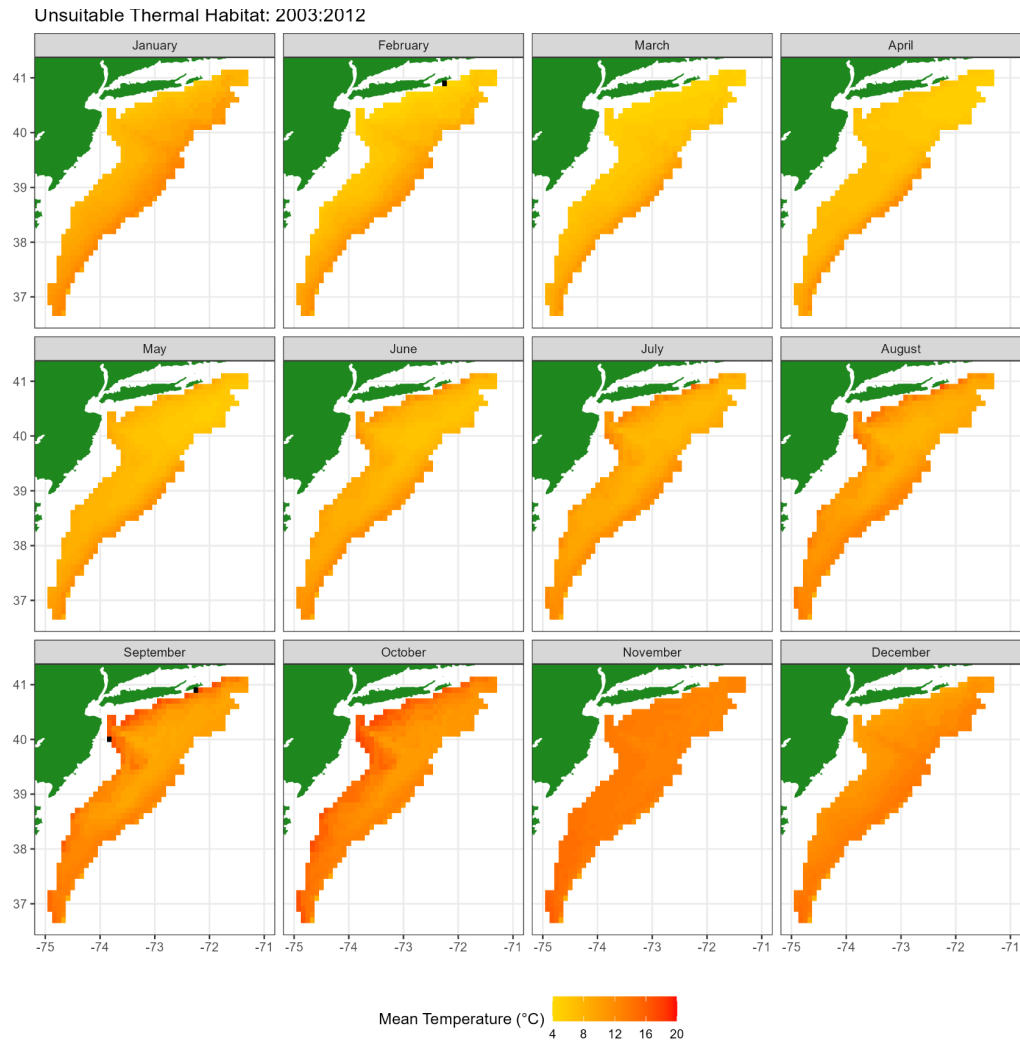


Figure 1.6B: Decadal unsuitable thermal habitat based on maximum monthly bottom temperature from 2003 to 2012 using GLORYS reanalysis. Black cells indicate areas with unsuitable thermal habitat.

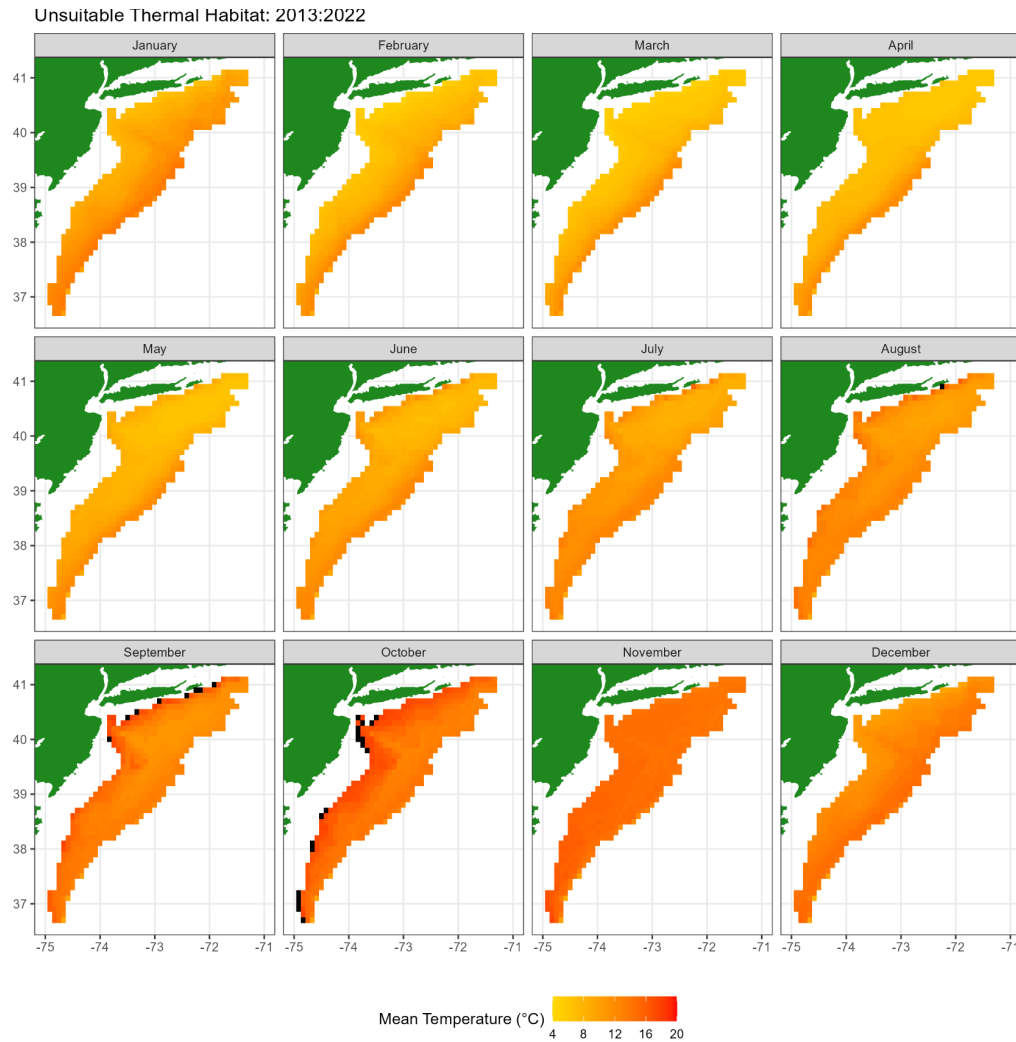


Figure 1.6C Decadal unsuitable thermal habitat based on maximum monthly bottom temperature from 2013 to 2022 using GLORYS reanalysis. Black cells indicate areas with unsuitable thermal habitat.

Abstracting from the particular geography of the MAB, it is possible to characterize BTH as a function of latitude and depth. Generally, shallower depths from 20-35°N area from August through October are outside of sea scallop's BTH (Figure 1.7). While mean monthly bottom temperature is warmer in shallower areas from August through October, none reach the threshold to be considered UTH (Figure 1.8). This indicates that while there may be localized areas of UTH in certain years and months, it is not a long-term characteristic of any regions within the SAMS areas.

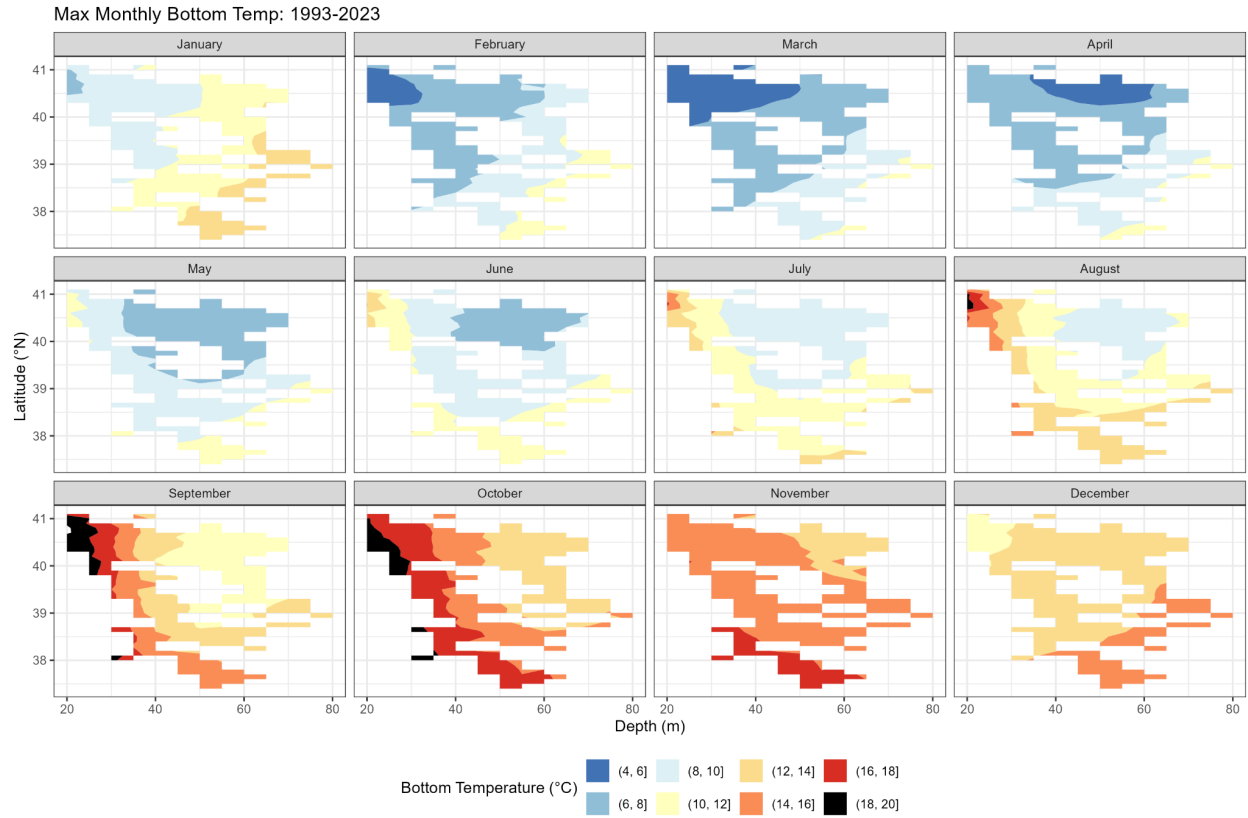


Figure 1.7: Baseline thermal habitat within the SAMS areas based on maximum monthly temperature from 1993 to 2023 as a function of latitude and depth bins. Black regions indicate areas with unsuitable thermal habitat.

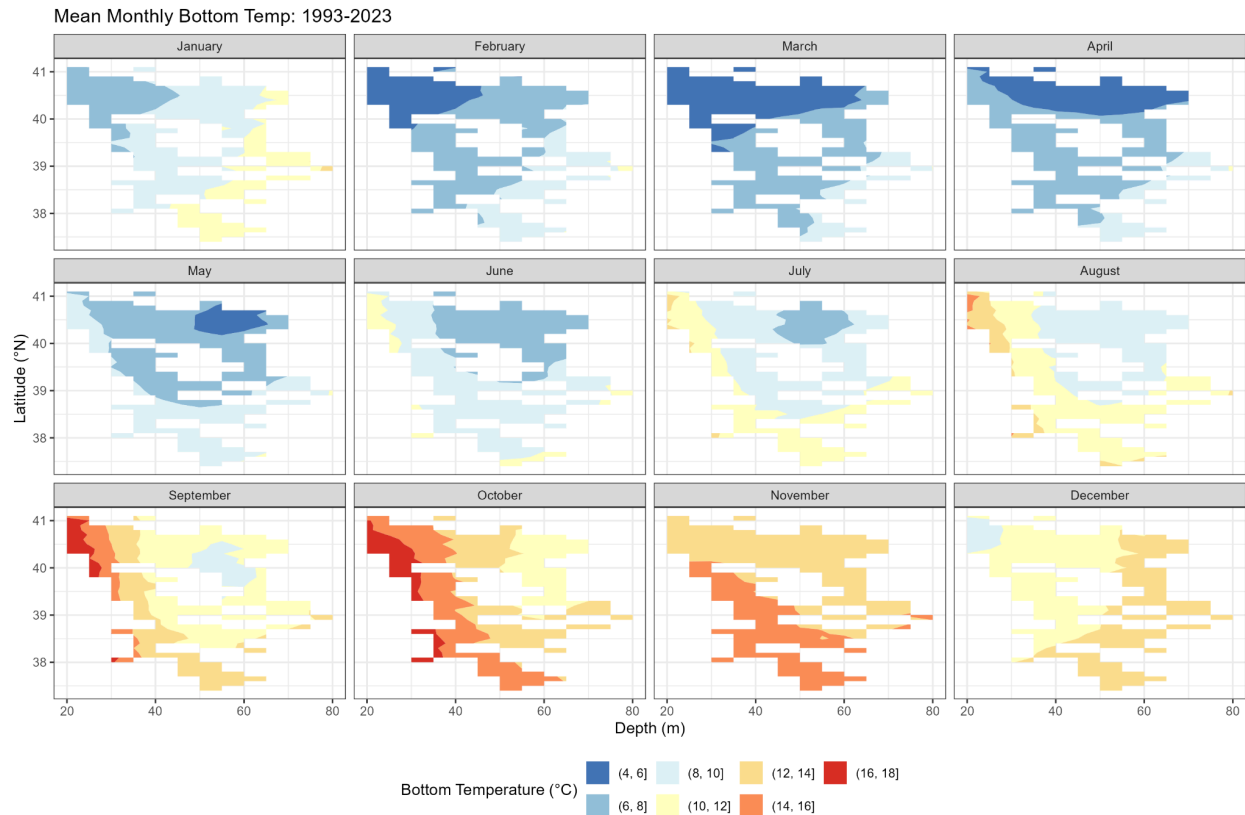


Figure 1.8: Unsuitable thermal habitat within the SAMS areas based on mean monthly temperature from 1993 to 2023 as a function of latitude and depth bins. Black regions indicate areas with unsuitable thermal habitat.

3.2. Heat stress event duration

Whereas BTH and UTH represent multi-annual metrics of thermal habitat, heat stress event duration (HSD) is a metric for a single year. Despite the years 1993 to 2012 having minimal areas of UTH, within a year GLORYS cells had between 31 and 95 days exceeding 18°C (Figures 1.9A and 1.9B). While most of these were isolated along inshore areas off of Long Island and New Jersey, some years (1999, 2001, and 2008) saw more frequent unsuitable areas further onto the shelf. In the most recent decade (2013 to 2023), HSE were more expansive throughout many of the SAMS areas, particularly in 2014, 2015, 2019, and 2023 (Figure 1.9C). In 2019, nearly the entirety of the SAMS areas experienced some HSE, besides some of Long Island and Block Island areas. In 2023, in addition to inshore HSE, were also present on the shelf break with the intermediate areas unaffected. The increased occurrence of HSE in these recent years also came with an increase in HSD with an annual range of 48 to 98 days depending on the grid cell. The longer events were also not isolated towards single cells. For instance, in 2019, nearly all of the Delmarva and Virginia SAMS areas exceeded 18°C for over 50 days.

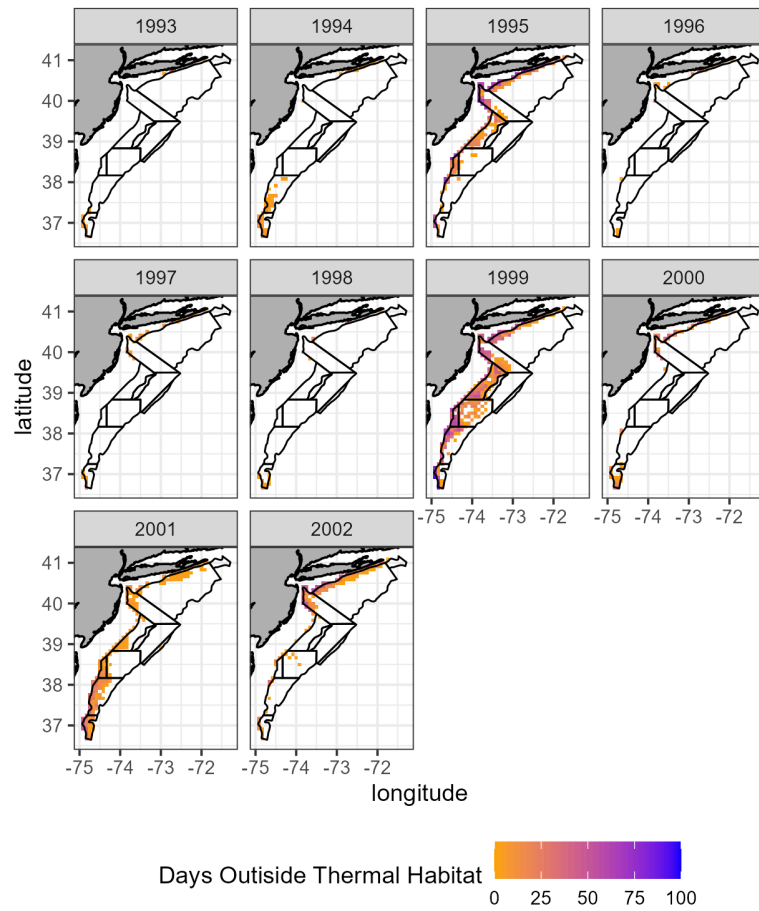


Figure 1.9A: The number of days where GLORYS bottom temperature exceeded 18°C from 1993 to 2002.

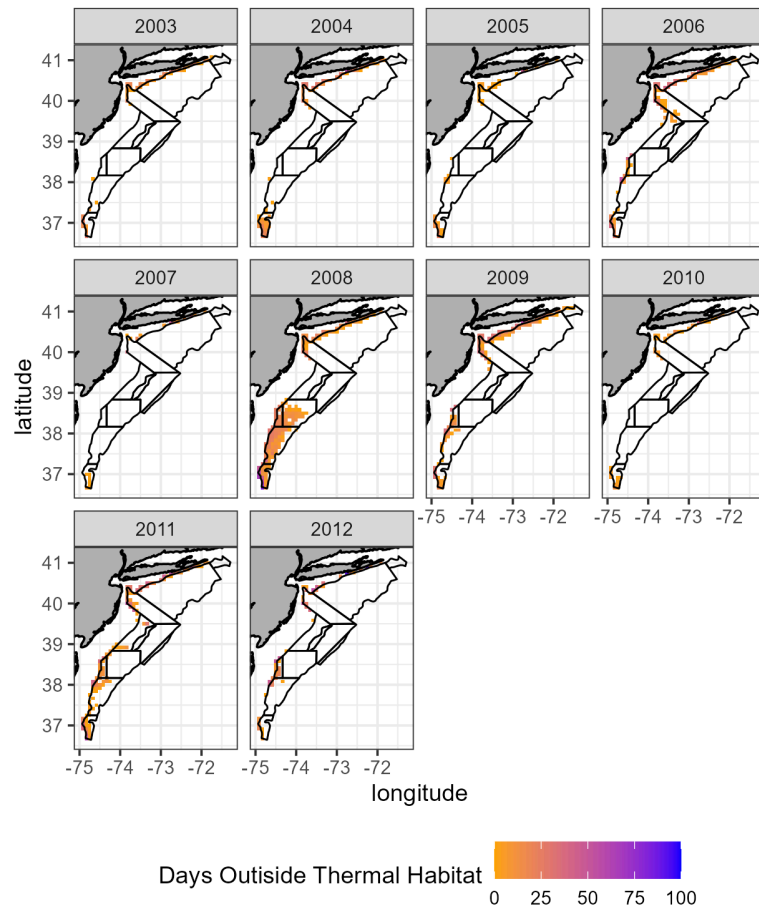


Figure 1.9B: The number of days where GLORYS bottom temperature exceeded 18°C from 2003 to 2012.

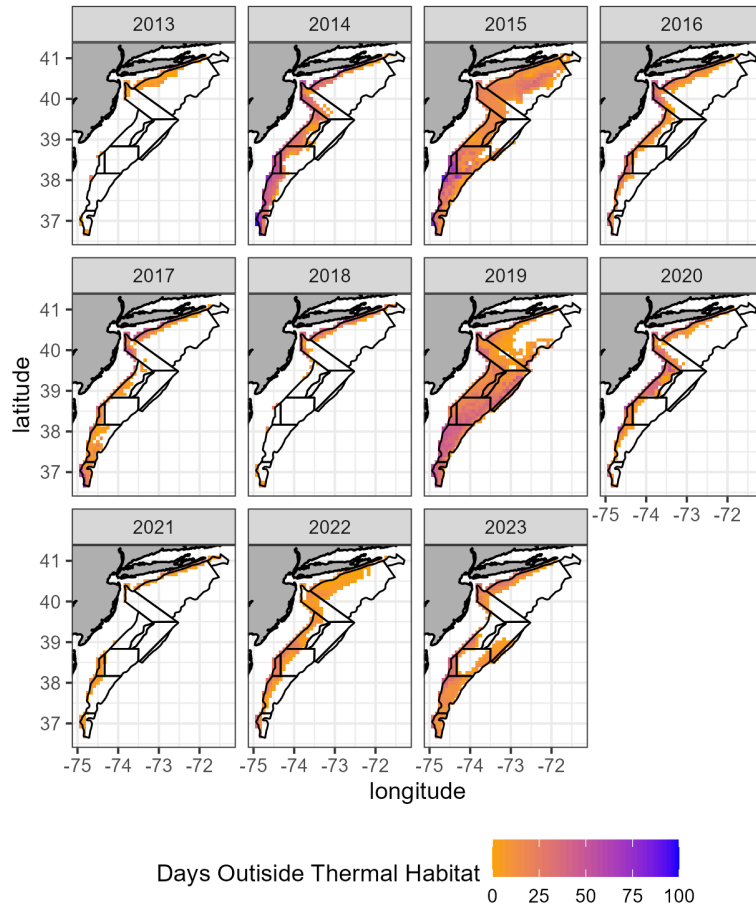


Figure 1.9C: The number of days where GLORYS bottom temperature exceeded 18°C from 2013 to 2023.

3.3. Heat Stress Area

The daily heat stress area (HSA) was calculated as the proportion of SAMS areas that experienced conditions greater than 18°C (Figure 1.10). The daily peak in HSA occurred in fall and ranged from less than 5% to 50% of the SAMS areas. Daily HSA has increased over this time ($R^2 = 0.017$, $p < 0.01$). The annual maximum HSA was also calculated as a measure of the most expansive HSE each year (Figure 1.11). This annual metric has also been increasing ($R^2 = 0.25$, $p = 0.002$), suggesting an increasing proportion of the SAMS area is experiencing some form of HSE for sea scallops within the year. Most of the HSE are occurring in the fall (October - December) and are increasing in both mean HSA ($R^2 = 0.19$, $p = 0.008$; Figure 1.12) and maximum HSA ($R^2 = 0.23$, $p = 0.003$; Figure 1.13).

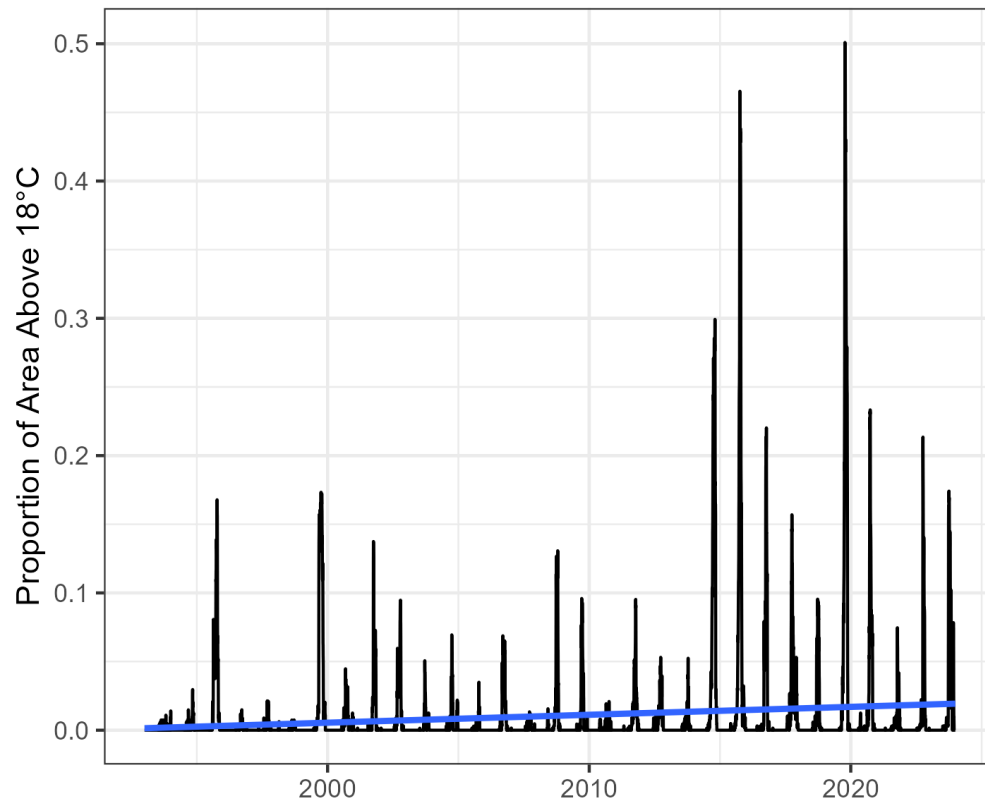


Figure 1.10: Daily heat stress area as proportion of SAMS areas. The blue line shows a linear regression.

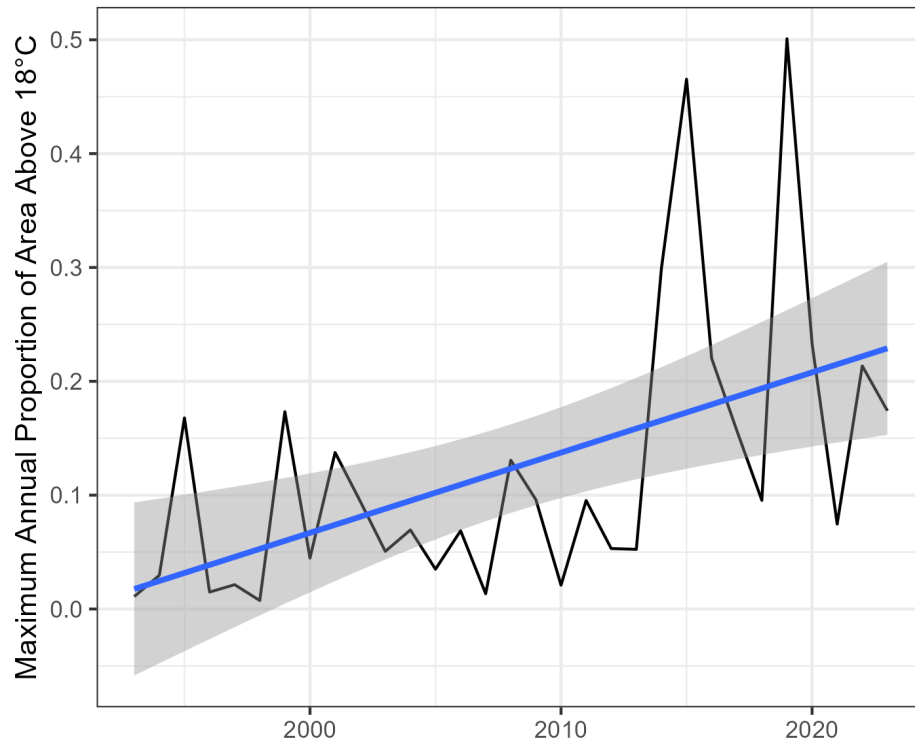


Figure 1.11: Annual maximum heat stress area as proportion of SAMS areas. The blue line shows a linear regression with grey shading showing the 95% confidence interval.

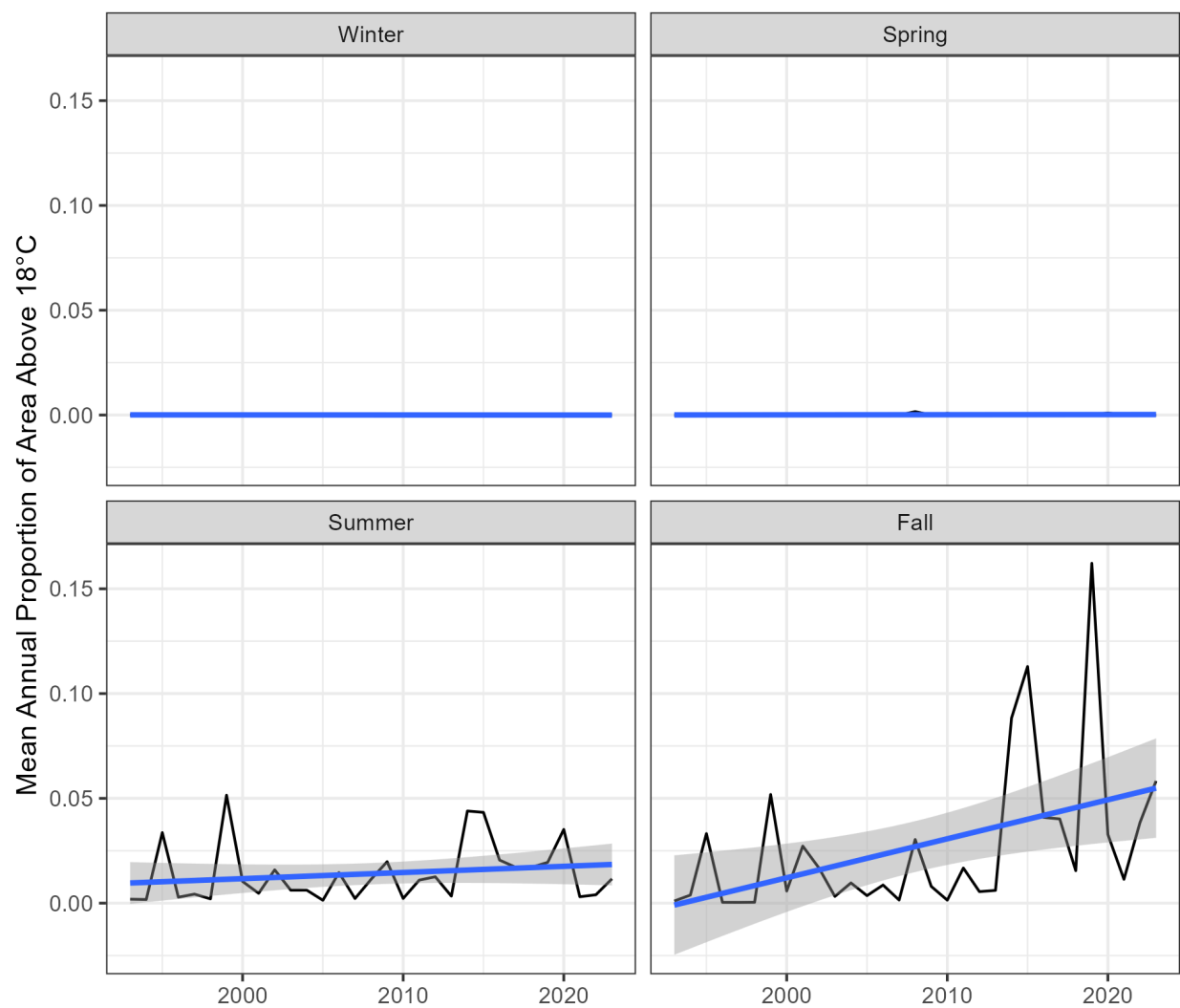


Figure 1.12: Mean seasonal heat stress area as proportion of SAMS areas. The blue lines show a linear regression with grey shading showing the 95% confidence interval. (Winter = January - March, Spring = April - June, Summer = August - September, Fall = October - December)

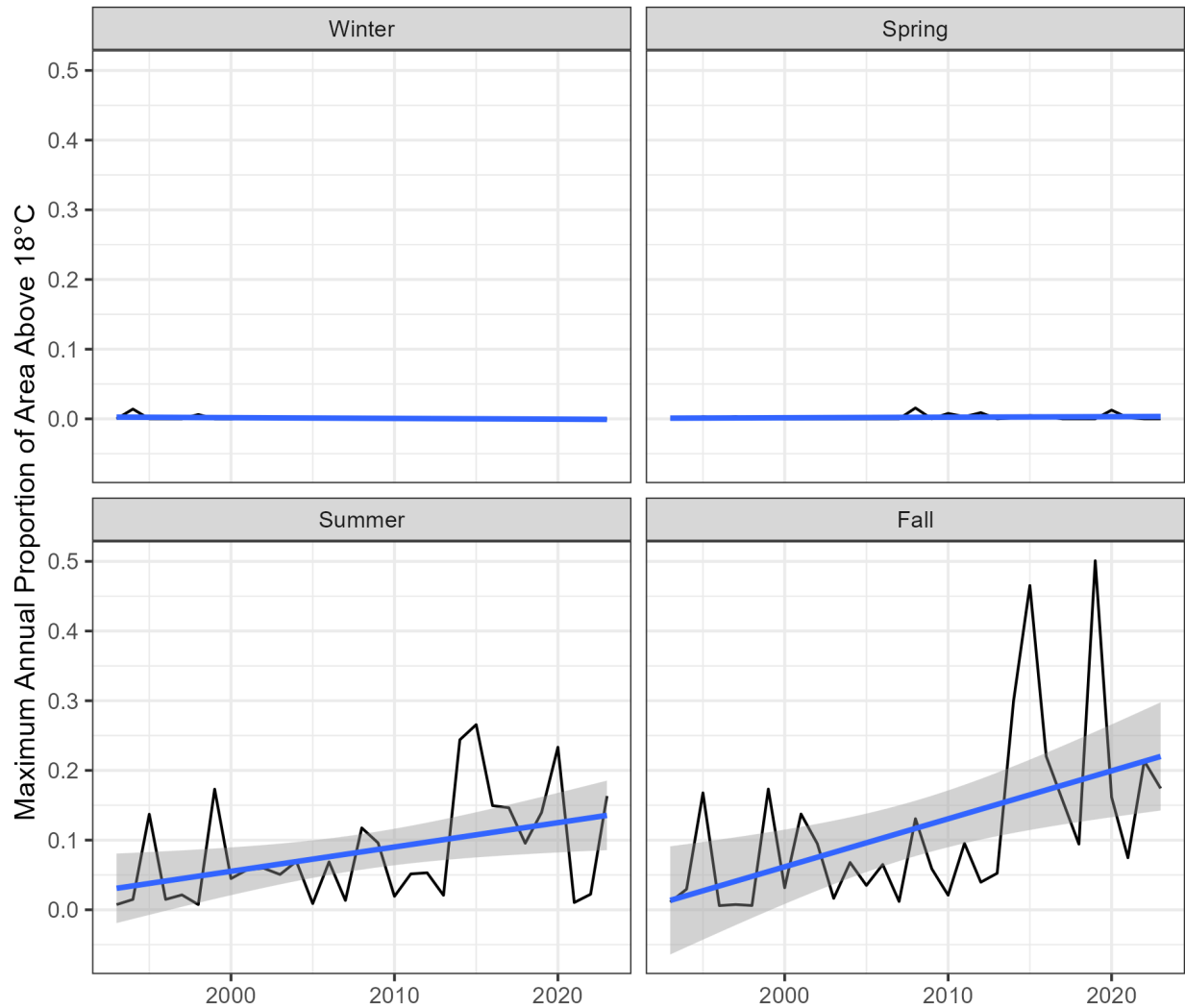


Figure 1.13: Maximum seasonal heat stress area as proportion of SAMS areas. The blue lines show a linear regression with grey shading showing the 95% confidence interval. (Winter = January - March, Spring = April - June, Summer = August - September, Fall = October - December)

3.4. Heat stress event intensity

Heat stress event intensity (HSI) is a daily measurement across all GLORYS cells within the SAMS areas and shows considerable variation within a year. Though HSA and HSE have been increasing since 1993, there is no significant trend in HSI ($R^2 = 0.015$, $p = 0.23$; Figure 1.14).

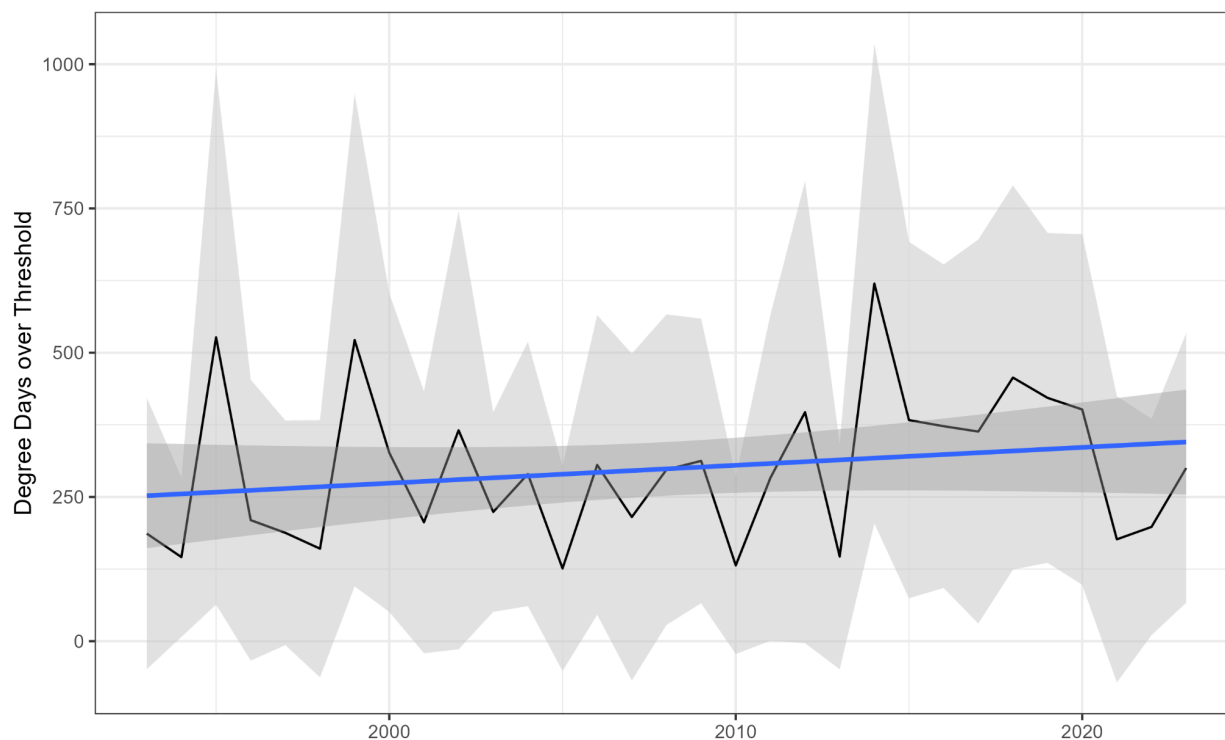


Figure 1.14: Heat stress intensity as calculated by degree days exceeding 18°C. The shaded area in light grey shows the range in daily degree days with the solid black line showing the annual mean. The blue line shows a linear regression with the dark grey shaded area as the 95% confidence interval.

4. Discussion

4.1. Patterns in baseline thermal habitat and heat stress events

Based on the analyses of GLORYS bottom temperature in the MAB, there is some evidence of long-term shrinking of baseline thermal habitat within the SAMS areas. In the last 10 years, much of the inshore portions of the SAMS areas are no longer within BTH. Despite this there has not been an increase in unsuitable thermal habitat to the same degree. While these changes in BTH do not extend further onto the shelf, there has been an observed increase in the extent and duration of heat stress events in the last decade. Depending on the year, offshore regions of the southern MAB experience HSE throughout entire SAMS areas (Elephant Trunk, Delmarva, and Virginia) for a month or more. Even though these locations are still within BTH conditions, they pose a high risk of heat-induced mortality or other physiological impacts to sea scallops. Most of the increase in HSA has been in late summer to early fall when bottom temperatures are generally higher. Due to the fact that such HSE are intermittent and not uniformly present with SAMS areas, the total HSE intensity shows no significant increase. This does not mean that HSE are not occurring, but rather that the areas that do experience HSE have a similar intensity over this time period.

4.2. Increases in risk to sea scallops

Since sea scallops are exotherms, their metabolism is susceptible to acute and prolonged temperature changes. The Northeast Climate Vulnerability Analysis (Hare et al., 2016) has identified Atlantic sea scallops as being at high risk and at high exposure to warming. If the expansive and long HSE observed in 2019 and 2023 continue, it may be worth considering an increased risk of climate exposure for sea scallops.

The decrease in BTH within the inshore SAMS areas poses a risk to the sea scallop, especially if BTH further contracts. The potential shifting of scallop distributions more offshore could result in increased travel cost for vessels. Since sea scallops are also constrained to the continental shelf in the MAB, these cross-shelf contractions pose an increased overlap with *Astropecten* predators on the shelf break. Since *Astropecten* are cold-limited species, warming bottom temperature can shift their distributions further onto the shelf, resulting in an increased predation risk for sea scallops.

The increases in HSE area and duration mean that sea scallops have been experiencing more thermal stress since the 1990s. An increase in thermal stress can manifest as changes in physiology (growth, gonad development, calcification, etc.; MacDonald & Thompson, 1986), but if severe enough could result in acute mortality events. The large mortality event seen in Elephant Trunk in 2022 does correspond to a moderate heat stress event in that region. However, the widespread HSE in 2019 and 2023 don't appear to correspond to any similar events seen in surveys.

There is also additional risk to sea scallops in the timing of these HSEs as they occur predominantly in the fall, during sea scallop spawning in the MAB (DuPaul et al., 1989). During spawning, sea scallop energy reserves are limited, and additional physiological stress could result in an amplified stress. If HSEs continue to coincide with spawning, this could result in additional mortality. If HSEs occur during gonad development, the additional metabolic demands could impact reproductive output of sea scallops, potentially impacting recruitment.

4.3. Other factors to consider

The initial goals of these analyses were to provide context for HSEs and to investigate the possibility of including an environmental covariate on natural mortality in the Atlantic sea scallop stock assessment model. However, at this point, these indicators alone do not directly correspond to all mortality events. It is likely that HSE indicators need to be assessed in conjunction with other environmental factors to be useful for an assessment model. Food availability during HSEs may be a potential added stressor, especially if metabolic demands are high. Further research should focus on developing indicators of benthic deposition of phytoplankton, particularly diatoms. Bottom oxygen concentrations can also have effects on sea scallop physiology, but since data are limited on the scale of SAMS areas, output from an oceanographic model would likely be needed. Additionally, GLORYs output is subject to error. At

the upper edge of an animal's thermal tolerance, an error of a single degree can be the difference between survival and mortality.

Once indicators for mortality are developed, they would need to be forecasted to be used in stock projections. Newly available annual and decadal forecasts are available for the Northwest Atlantic using NOAA's MOM6 model (Ross et al., 2024). Validating and distributing these outputs for use in stock assessments is ongoing.

4.4. Limitations of data

The entirety of this analysis is based on an evaluation of the GLORYS reanalysis product. While this product is the recommended global reanalysis for the northeast U.S. shelf (Castillo-Trujillo et al., 2023), it does not perform equally well in all areas. As a data assimilative model, observations are used when available to “nudge” the underlying oceanographic model. However, given the limited spatio-temporal bottom temperature observations, there is still a need to quantify model uncertainty if being used within a stock assessment model.

5. Conclusion

The concern that warming bottom temperatures will impact Atlantic sea scallop distributions, mortality, and productivity have led to investigations of patterns in thermal habitat and heat stress events using output from GLORYS reanalysis. The results of this analysis indicate that there is a contraction of baseline thermal habitat in the MAB, particularly in the inshore region of the SAMS areas. There has also been an increasing trend in the duration and spatial extent of heat stress events in September and October. A continued increase in these heat stress events pose serious risks for scallop mortality and physiology in the region, which in turn could have significant impacts to the fishery. Further investigation is needed to incorporate these environmental stressors into the stock assessment either directly into the assessment model or as additional risk factors. However, heat stress indices alone do not appear to predict all observed mortality events. This work will continue to explore other environmental stressors using satellite and oceanographic model output to improve the predictive capacity for future mortality events.

6. References

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